

A weighted-cycle obstruction to a conjectured surjectivity from the orthogonal group

Candidate counterexample to the conjecture in arXiv:2408.04346

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Status: candidate counterexample, likely valid. For

$$\Phi_n(O)_{ij}^2 = \frac{1}{2}(O_{ij}^2 + O_{ji}^2), \quad O \in \mathrm{SO}(n),$$

the source conjectures surjectivity onto the symmetric nonnegative matrices whose Hadamard square is doubly stochastic. This is false already for $n = 6$. A doubly stochastic matrix supported on a 6-cycle with alternating edge weights $1/3, 2/3$ cannot be the symmetrized entrywise square of any orthogonal matrix. The same construction disproves surjectivity for every $n \geq 6$.

1 The source conjecture

The source defines a weight matrix $\Theta = \Phi_n(O)$ from a special orthogonal matrix by

$$\Theta_{ij} = \sqrt{\frac{1}{2}(O_{ij}^2 + O_{ji}^2)}.$$

It observes that Θ is symmetric and that $P = \Theta \circ \Theta$ is doubly stochastic, then conjectures that this map is surjective onto all symmetric nonnegative matrices having doubly stochastic Hadamard square [1, p. 5]. The complete source page is reproduced below.

Here and in the sequel, I denotes $n \times n$ identity matrix. Moreover, we shall always denote the Stieltjes transform of the semicircle law by

$$s(z) = \frac{1}{2\pi} \int_{-2}^2 \frac{\sqrt{4-x^2}}{x-z} dx.$$

The results in [GNT15] in particular involve the convergence of the expected Stieltjes transforms $\mathbb{E}_X s_\Theta(z) \rightarrow s(z)$ as $n \rightarrow \infty$. Moreover, for independent X_{ij} with doubly stochastic variance profile rates of convergence for $\Delta_\Theta := \sup_x |\mathbb{E}_X F_\Theta(x) - G(x)| \rightarrow 0$ and $\mathbb{E}_X s_\Theta(z) \rightarrow s(z)$ as $n \rightarrow \infty$ were derived together with local laws in [EYY12] under stronger moment conditions and more restrictive conditions on the weights, requiring $n\Theta_{ij}^2$ to be bounded above and below by positive constants. Our model of weights will be more general, only assuming (4).

We would like to study the concentration of measure phenomenon for $\Theta \mapsto \mathbb{E}_X s_\Theta$ in the spirit of Sudakov for our model. Since the set of doubly stochastic square weights $P_{ij} = \Theta_{ij}^2$ is a convex set, [CDS10] modelled it via a uniform distribution on it. However, since we intend to use log-Sobolev techniques, the latter turns out to be inconvenient. We therefore suggest the following probability model, where Θ is a functional on a space better suited to our purpose.

Let $\text{SO}(n)$ denote the special orthogonal group of dimension n , which we endow with its Haar measure $\mu_{\text{SO}(n)}$, and consider the mapping

$$(2.1) \quad O \mapsto \Theta = (\Theta_{ij})_{ij}, \quad \text{where } \Theta_{ij} := \sqrt{\frac{1}{2}(O_{ij}^2 + O_{ji}^2)}.$$

Obviously, Θ is symmetric, and its Hadamard square is doubly stochastic by the orthogonality of O . In fact, it seems reasonable to conjecture that (2.1) defines a surjective mapping into the space of the symmetric matrices with non-negative entries and doubly stochastic Hadamard squares. Therefore, we equip the space of the weight matrices with the measure induced by (2.1). In particular, (4) is satisfied automatically for this model (cf. e.g. Lemma 2.5 below).

2.1.1. Concentration results. In studying a local semicircular law, we first establish sub-Gaussian concentration of $\mathbb{E}_X s_\Theta(z)$ around the expected Stieltjes transform $\mathbb{E} s_\Theta(z)$. The key observation is to identify $\mathbb{E}_X s_\Theta(z)$ as a Lipschitz function of $O \in \text{SO}(n)$.

Before stating the result, let us fix some notation which we will use all over this section. As stated before, $\mathbb{E}_X$ denotes taking the expectation with respect to the matrix X , and $\mathbb{E}_{\text{SO}(n)}$ denotes taking the expectation with respect to O which is distributed according to the Haar probability measure on $\text{SO}(n)$. In particular, we require X and O to be independent, so that $\mathbb{E} = \mathbb{E}_X \mathbb{E}_{\text{SO}(n)}$. Moreover, given any $z \in \mathbb{C}$, we set $v := \Im(z)$.

Proposition 2.1. *Let $n \geq 3$ and $z \in \mathbb{C} \setminus \mathbb{R}$. Then, it holds that*

$$\mu_{\text{SO}(n)}(|\mathbb{E}_X s_\Theta(z) - \mathbb{E} s_\Theta(z)| \geq t) \leq 2 \exp \left\{ -\frac{|v|^4 n^2}{768} t^2 \right\}$$

for any $t \geq 0$. Furthermore, if $|X_{ij}| \leq K$ for any i, j and some $K > 0$, it holds that

$$\mu_{\text{SO}(n)}(|s_\Theta(z) - \mathbb{E}_{\text{SO}(n)} s_\Theta(z)| \geq t) \leq 2 \exp \left\{ -\frac{|v|^4 n^2}{768 K^2} t^2 \right\}$$

for any $t \geq 0$.

Equivalently, the conjecture says that every symmetric doubly stochastic matrix P can be written

$$P_{ij} = \frac{1}{2}(O_{ij}^2 + O_{ji}^2) \quad (1 \leq i, j \leq n) \quad (1)$$

for some $O \in \text{SO}(n)$.

2 Proof intuition

Zeros in P are rigid: because the summands in (1) are nonnegative, $P_{ij} = 0$ forces both $O_{ij} = 0$ and $O_{ji} = 0$. We use a sparse P whose support is an undirected cycle. On a cycle of length at least five, two rows at distance two have only one possible common nonzero column. Orthogonality forces one of those two entries to vanish. Column normalization then forces the other to have modulus one.

Repeating this around the cycle makes O a signed permutation, whose symmetrized squared entries can only be 0, $1/2$, or 1. Fractional cycle weights such as $1/3$ and $2/3$ are therefore impossible.

3 Cycle-support rigidity

Write indices modulo m , and let C_m have edges $\{i, i+1\}$.

Lemma 1. *Let $m \geq 5$. If $O \in O(m)$ satisfies*

$$O_{ij} = 0 \quad \text{whenever} \quad \{i, j\} \notin E(C_m),$$

including $O_{ii} = 0$, then O is a signed permutation matrix.

Proof. Fix a column j . Its only potentially nonzero entries are $O_{j-1,j}$ and $O_{j+1,j}$. Rows $j-1$ and $j+1$ are supported, respectively, in

$$\{j-2, j\} \quad \text{and} \quad \{j, j+2\}.$$

Since $m \geq 5$, these two supports meet only in column j . Orthogonality of the two rows therefore gives

$$O_{j-1,j}O_{j+1,j} = 0.$$

On the other hand, column j has norm one, so

$$O_{j-1,j}^2 + O_{j+1,j}^2 = 1.$$

Exactly one entry in the column is thus ± 1 , and the other is zero. This holds for every column. Orthogonality (or row normalization) then gives exactly one ± 1 in every row as well. $\square$

4 The counterexample

Theorem 2. *The map $\Phi_6 : SO(6) \rightarrow \mathbb{R}^{6 \times 6}$ in the source is not surjective onto the symmetric nonnegative matrices whose Hadamard squares are doubly stochastic. In fact, the target matrix $\Theta = P^{\circ 1/2}$ with*

$$P = \frac{1}{3} \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

has no preimage even in $O(6)$.

Proof. The matrix P is symmetric and nonnegative. Every row contains one entry $1/3$ and one entry $2/3$, so every row and column sums to one. Thus $\Theta = P^{\circ 1/2}$ belongs to the conjectured target.

Suppose (1) held for an orthogonal O . Whenever $P_{ij} = 0$, nonnegativity in (1) forces $O_{ij} = O_{ji} = 0$. Hence O is supported on the displayed 6-cycle and has zero diagonal. By Lemma 1, O is a signed permutation matrix. Therefore $(O_{ij}^2, O_{ji}^2) \in \{0, 1\}^2$, and (1) forces

$$P_{ij} \in \{0, \frac{1}{2}, 1\} \quad \text{for all } i, j.$$

But every nonzero entry displayed in P is $1/3$ or $2/3$, a contradiction. $\square$

Corollary 3. *For every even $m \geq 6$, alternating edge weights $a, 1 - a$ on C_m , with*

$$0 < a < 1, \quad a \notin \{\frac{1}{2}\},$$

give a one-parameter family of counterexamples. Surjectivity also fails in every dimension $n \geq 6$.

Proof. On an even cycle the alternating weights make every row sum one. Lemma 1 again forces any putative preimage to be a signed permutation, while its symmetrized square has entries only in $\{0, 1/2, 1\}$. Since both a and $1 - a$ lie strictly between zero and one and differ from $1/2$, no preimage exists.

For arbitrary $n \geq 6$, take the direct sum of the displayed 6×6 matrix with I_{n-6} . It is symmetric doubly stochastic. The zero off-block entries would force any preimage to preserve the two blocks, leaving the impossible 6-dimensional block. $\square$

5 Checks, upgrade attempts, and scope

Independent finite check. Exact rational arithmetic verifies symmetry, nonnegativity, and all row sums of P . Enumerating the permutation matrices supported on C_6 finds four; every symmetrized square has entries in $\{0, 1/2, 1\}$, and none equals P . The packet-local script `code/verify_weighted_cycle.py` is confirmatory; the proof above is exact and does not rely on enumeration.

Deep upgrades. The first construction targeted only $n = 6$. The support argument upgrades it to a rigidity lemma for every cycle of length at least five, a one-parameter counterexample family in every even dimension at least six, and, by block embedding, failure in every dimension $n \geq 6$. The proof does not determine whether surjectivity holds for $n = 3, 4, 5$; those smaller cases are outside the counterexample claim.

Novelty check. Bounded primary-source searches through 11 August 2026 covered the exact formula, the source title, symmetrized orthostochastic matrices, and symmetric bistochastic matrices. They recovered the source and nearby work on orthostochastic and symmetric-bistochastic subclasses, but no later primary source stating this counterexample or resolving the conjecture. The search was bounded, so novelty confidence is moderate.

Human-review recommendation. Check the zero-propagation step, the one-column intersection in Lemma 1, and the direct-sum extension.

References

- [1] F. Götze and H. Sambale, *Concentration of measure on spheres and related manifolds*, [arXiv:2408.04346](https://arxiv.org/abs/2408.04346).